ARIF AHMAD KHAN

Curriculum Vitae | Artificial Intelligence and Agentic Systems

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RESEARCH INTERESTS

Agentic reasoning and long horizon autonomy in large language models, including planning, tool use, and self correction across extended task execution; memory architectures and context management for persistent, stateful agents; multi agent coordination protocols and emergent collaborative behavior; inference time reasoning and test time compute scaling; evaluation and interpretability of agentic systems, with emphasis on verifiable reasoning traces, reward model alignment, and reliability under real world deployment conditions.

PROFILE

AI Engineer specializing in agentic systems, retrieval augmented generation (RAG), and large language model application backends. Designs and deploys production multi agent architectures using LangGraph and LangChain, with scalable FastAPI services grounded in vector search. Has independently built and deployed seven end to end agentic AI systems spanning legal document analysis, sales automation, conversational ordering, and IT operations, several of which operate autonomously in live production environments. Seeking graduate study in Artificial Intelligence or Computer Science to strengthen theoretical foundations in machine learning and multi agent systems research.

EDUCATION

B.S., Software Engineering, University of Malakand, Pakistan *2021 to 2025*

- Cumulative GPA: 3.65 / 4.00
- Relevant coursework: Linear Algebra, Discrete Structures, Stochastic Processes, Operations Research, Formal Methods in Software Engineering, Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Probability and Statistics.

CERTIFICATIONS AND ADVANCED COURSEWORK

Courses taught by Andrew Ng and the DeepLearning.AI team through Coursera. Each underlined title opens its verifiable credential.

Machine Learning Specialization, DeepLearning.AI and Stanford Online *Jul 2023*

- Completed: Supervised Machine Learning: Regression and Classification (Mar 2023); Advanced Learning Algorithms (May 2023); Unsupervised Learning, Recommenders, Reinforcement Learning (Jul 2023). Specialization certificate.
- Covers regression, neural networks, decision trees, clustering, recommenders, and reinforcement learning.

Deep Learning Specialization, DeepLearning.AI *Jul to Aug 2023*

- Four courses: Neural Networks and Deep Learning (Jul 2023); Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization (Jul 2023); Structuring Machine Learning Projects (Jul 2023); Convolutional Neural Networks (Aug 2023).

Advanced TensorFlow Coursework, DeepLearning.AI, taught by Laurence Moroney of Google *Jul 2024*

- Custom Models, Layers, and Loss Functions with TensorFlow (Jul 2024); Custom and Distributed Training with TensorFlow (Jul 2024).

Agentic AI Training *2025*

- Understanding Agentic AI, AgentAcademy; Fundamentals of Agents, Hugging Face.

TECHNICAL SKILLS

Agentic AI and LLMs: LangGraph, LangChain, RAG and vector search, Ollama, Hugging Face Transformers, Groq, Cerebras, OpenAI, Anthropic Claude, Google Gemini, prompt engineering

Programming and Data: Python, SQL, Pandas, NumPy, R (basic)

Machine Learning: TensorFlow, Keras, Scikit-learn, OpenCV

Backend, Cloud and DevOps: FastAPI, AWS, Microsoft Azure, Vercel, Docker, Git, Azure DevOps

Databases: Pinecone, Weaviate, FAISS (vector); PostgreSQL, MySQL, Supabase (relational)

Web Development: React, Node.js, TypeScript, JavaScript

Automation Platforms: n8n, Zapier, Make, Microsoft Power Automate and Power Platform

PROFESSIONAL EXPERIENCE

AI Engineer, Int. Dublin Tax Company Ltd. *Present*

- Build and manage AI driven automation workflows on Microsoft Power Automate and the Power Platform, reducing manual processing across internal tax and accounting operations.
- Develop and ship AI agents, workflows, and custom copilots supporting internal tax and financial operations (implementation details confidential due to the sensitivity of client financial data).
- Develop AI powered virtual agents and support enterprise cloud infrastructure on Microsoft Azure.

Founder and AI Engineer, AgentCraft AI Automations *2025 to Present*

- Founded and operate an AI automation studio delivering custom agentic systems, RAG assistants, and workflow automation for client businesses.
- Own the full delivery cycle across discovery, architecture, implementation, testing, and deployment, using LangGraph, FastAPI, and workflow integration platforms.

Computer Vision Intern, ITSOLERA Pvt. Ltd. *Jul 2024 to Oct 2024*

- Developed real time pedestrian detection systems in Python and OpenCV.
- Built object detection pipelines robust to low light, rain, and fog conditions.
- Contributed to a 3D medical imaging reconstruction system and to a cardiovascular disease risk assessment project based on AI driven medical image analysis.

FINAL YEAR PROJECT AND SELECTED TECHNICAL PROJECTS

Spatial AI Vision Assistant for the Visually Impaired *(Final Year Project; Computer Vision, Edge AI, FastAPI, Groq)*

- Built a real time hazard detection pipeline combining quantized YOLO models with Depth Anything V2 INT8 ONNX monocular depth estimation to detect obstacles and ground drop offs across left, center, and right spatial corridors.
- Optimized CPU inference with OpenVINO and ONNX Runtime to reach 92 FPS (under 15 ms latency) over WebSockets; deployed on Vercel as a WCAG AAA compliant PWA with Groq scene reasoning and a "Silent Until Danger" alert engine. *Live: ai-vision-assistant-fyp.vercel.app | GitHub: ahmadarif238/ai-vision-assistant-fyp*

ContractIQ: Legal Agreement Analysis *(LangGraph, Agentic AI)*

- Autonomous multi agent system that analyzes, verifies, and benchmarks complex legal agreements.
- Designed a zero hallucination architecture separating risk reasoning from clause and lifecycle extraction, keeping model outputs traceable to source text and reducing false positives.

Client Hunting and Lead Generation Agent *(Python, Web Scraping, LLM Personalization)*

- Production agent that scrapes business listings, evaluates each prospect's web presence for automation opportunities, and drafts personalized outreach based on identified gaps.
- Operates autonomously in daily production, handling approximately 80 personalized outreach messages per day with automated follow up and reply routing.

WhatsApp Conversational Ordering Agent *(LLM, Speech to Text, React, Node.js)*

- Production conversational agent handling multilingual customer interaction in English, Urdu, and Roman Urdu, including transcription of voice message input, through to order completion.
- Built the companion operations dashboard used by staff to confirm orders and relay real time status updates to customers.

VivaGraph AI: Oral Examination Simulator *(LangGraph, Agentic AI)*

- Multi agent platform with a seven agent cognitive core simulating oral examinations.
- Integrates real time voice analysis, filler word detection, and dynamic difficulty adjustment, with each agent specializing while sharing a unified session state.

SalesAI: B2B Sales Automation *(FastAPI, Agentic AI)*

- Orchestrates four specialized sub agents automating B2B lead generation, scoring, and engagement, reducing manual research time by 60%.

AI Operations Copilot *(LangGraph, RAG)*

- Multi agent RAG system for IT operations covering policy question answering, log monitoring, and automated ticket generation.

LANGUAGES

English (Duolingo English Test 130/160, CEFR C1 Advanced) | Urdu (native) | Pashto (native)